TEXAS A&M UNIVERSITY

Course Syllabus

Course Information

Course Number: ECEN689
Course Title: Special Topics Multimodal Computer Vision - Principles and Applications
Lecture Time: TBD
Lecture Location: TBD

Labs:
Credit Hours: 3 (3 hours of lecture/discussion per week; students are expected to spend an additional ~3 hours per week on readings, assignments, and project work outside class)

Instructor Details

Instructor: Zhiwen Fan
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Office Hours: TBD

Course Description

This course examines the motivations, algorithms, and models in computer vision that integrate multiple modalities (e.g., images, text, LiDAR). Students will explore image formation, 3D vision, representation learning, and sensor fusion, with applications in robotics, autonomous driving, and the metaverse. Emphasis is placed on core principles—such as image recognition, multi-view geometry, and self-supervised learning—and on hands-on labs where students build, fine-tune, and deploy multimodal vision models for tasks like text-to-image generation, 3D reconstruction, and robotic perception.

Course Prerequisites

Prerequisites:

1. ECEN 447 or ECEN 642

Special Course Designation

This graduate-level course includes advanced labs focused on research paper reading and experimental implementation. The final project shall center on a specific research problem, requiring students to implement and evaluate a state-of-the-art algorithm in multimodal computer vision.

Course Learning Outcomes

Learning Outcomes:

Upon the completion of the course, students will:

- Analyze and apply deep learning architectures (e.g., convolutional neural networks, transformers) for vision tasks.
- Explain fundamental 3D vision principles, including multi-view geometry, Neural Radiance Fields (NeRFs), and 3D Gaussian splatting.
- Develop and evaluate generative vision models that produce images, video, or 3D content from textual input.
- Integrate multimodal vision techniques into applications such as robotics perception, autonomous driving, or metaverse/AR/VR scenarios.
- Design, implement, and assess algorithms to solve existing 3D vision challenges, with appropriate experiments and evaluation metrics.

Textbook and/or Resource Materials

Textbook – no required text. Recommended Resources: Course notes and on-line design tutorials will be provided.

Grading Policy

Course grades will be determined from:

- Participation (5%)

Regular attendance, active engagement in discussions, and thoughtful questions.

- Individual Paper Presentations (20%)
  - Each student presents one or more papers on a selected topic.
  - Graded on clarity, depth of understanding, and quality of communication.
- Midterm Literature Review Report (20%)
  - Expand upon a presented paper or chosen research direction by surveying 6–10 related papers.
  - Write a concise report (at least 4 pages, excluding references) using the NeurIPS 2025 LaTeX template.
  - The report should include:
    - ✓ Introduction and motivation
    - ✓ Background overview
    - ✓ Descriptions of key algorithms, their interconnections, and salient experimental findings
    - ✓ Clear organization and synthesis of insights
- Final Team Project (55%)
  - Proposal Presentation (10%)
    - ✓ Brief in-class presentation of the chosen research problem, including motivation, problem setup, baseline methods, proposed approach, and expected outcomes.
    - ✓ Must align with course topics.
    - ✓ Graded on clarity, rationale, and preparedness.
  - Final Project Presentation (20%)
    - ✓ Detailed presentation covering problem statement, methodology, experiments, results, and analysis.
    - ✓ Graded on clarity, depth of analysis, and quality of delivery.
  - Final Project Report (25%)
    - ✓ A written report (up to 6 pages, excluding references; supplementary material encouraged) in NeurIPS 2025 LaTeX format.
    - ✓ Should include: introduction, related work, approach, experimental design and results, discussion, and conclusion.
    - ✓ Aim for workshop-level quality at a top-tier venue.
    - ✓ Graded on novelty, completeness, rigor of experiments, clarity of writing, and demonstration of effort.

Grading Scale:

A = 90-100

B = 80-89

C = 70-79

D = 60-69

F = <60

Late Work Policy

⩽ 2 late work will be accepted;

Course lecture topics, major assignments

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Week	Topic	Assignments

1	Introduction to Vision and Multimodal Learning	
2	Image Processing and Neural Networks	- Read foundational chapters/papers on CNNs and Vision Transformers - Assignment: implement a basic CNN classifier or experiment with a pretrained ViT on a small dataset
3	Principles in 3D Reconstruction: Multi-View Geometry	- Read on multi-view geometry (e.g., epipolar geometry) - Assignment: implement or use an SfM code to estimate camera pose / depth from image pairs
4	Neural Radiance Fields and 3D Gaussian Splatting	- Read NeRF paper and a Gaussian splatting reference - Assignment: run an existing NeRF or Gaussian-splatting demo; analyze quality vs. compute/time
5	Guest Lecture: Scene Understanding and Robotics	
6	Single-View to 3D	
7	2D Image Generation	- Read on generative models (GANs or diffusion) for 2D image synthesis - Assignment: generate images via a public model or small training run; analyze strengths/limitations
8	End-to-End 3D Reconstruction and Generation	- Read on integrated pipelines that combine reconstruction and generative components - Assignment: prototype or evaluate a pipeline assembling reconstruction → generative step - Midterm Literature Review Report Due: survey 6–10 papers on your chosen topic (4+ pages, NeurIPS style)
9	Metaverse and Real-Time Systems	
10	Self-Supervised Pre-training	- Read on contrastive learning, masked modeling for vision (and multimodal if available) - Assignment: visualizing and analyzing several pretrained self-supervised models (e.g., via PCA); report observations
11	Large Language Models and Vision-Language Models	- Read on CLIP, Flamingo, ViLT or related VLM architectures - Assignment: experiment with a vision-language model (e.g., prompt-based retrieval or VQA demo) and summarize results
12	3D-Aware Vision-Language Models for Autonomous Systems	- Read on 3D-aware multimodal models applied to robotics/autonomous systems - Assignment: evaluate or prototype a 3D-aware VLM component for a simple scenario (e.g., spatial navigation in a 3D scene)
13	Autonomous Driving	

14	Final Project Presentations	- Students present final work: problem statement, implemented state-of-the-art algorithm, experiments, results, and critical analysis - Submit final project report (workshop-level, NeurIPS-style)

University Policies

Attendance Policy

The university views class attendance and participation as an individual student responsibility. Students are expected to attend class and to complete all assignments.

Please refer to Student Rule 7 in its entirety for information about excused absences, including definitions, and related documentation and timelines.

Makeup Work Policy

Students will be excused from attending class on the day of a graded activity or when attendance contributes to a student's grade, for the reasons stated in Student Rule 7, or other reason deemed appropriate by the instructor.

Please refer to Student Rule 7 in its entirety for information about makeup work, including definitions, and related documentation and timelines.

Absences related to Title IX of the Education Amendments of 1972 may necessitate a period of more than 30 days for make-up work, and the timeframe for make-up work should be agreed upon by the student and instructor" (Student Rule 7, Section 7.4.1).

"The instructor is under no obligation to provide an opportunity for the student to make up work missed because of an unexcused absence" (Student Rule 7, Section 7.4.2).

Students who request an excused absence are expected to uphold the Aggie Honor Code and Student Conduct Code. (See Student Rule 24.)

Academic Integrity Statement and Policy

"An Aggie does not lie, cheat or steal, or tolerate those who do."

"Texas A&M University students are responsible for authenticating all work submitted to an instructor. If asked, students must be able to produce proof that the item submitted is indeed the work of that student. Students must keep appropriate records at all times. The inability to authenticate one's work, should the instructor request it, may be sufficient grounds to initiate an academic misconduct case" (Section 20.1.2.3, Student Rule 20).

You can learn more about the Aggie Honor System Office Rules and Procedures, academic integrity, and your rights and responsibilities at aggiehonor.tamu.edu.

Americans with Disabilities Act (ADA) Policy

Texas A&M University is committed to providing equitable access to learning opportunities for all students. If you experience barriers to your education due to a disability or think you may have a disability, please contact Disability Resources in the Student Services Building or at (979) 845-1637 or visit disability.tamu.edu. Disabilities may include, but are not limited to attentional, learning, mental health, sensory, physical, or chronic health conditions. All students are encouraged to discuss their disability related needs with Disability Resources and their instructors as soon as possible.

Title IX and Statement on Limits to Confidentiality

Texas A&M University is committed to fostering a learning environment that is safe and productive for all. University policies and federal and state laws prohibit gender-based discrimination and sexual harassment, including sexual assault, sexual exploitation, domestic violence, dating violence, and stalking.

With the exception of some medical and mental health providers, all university employees (including full and part-time faculty, staff, paid graduate assistants, student workers, etc.) are Mandatory Reporters and must report to the Title IX Office if the employee experiences, observes, or becomes aware of an incident that meets the following conditions (see University Rule 08.01.01.M1):

- The incident is reasonably believed to be discrimination or harassment.
- The incident is alleged to have been committed by or against a person who, at the time of the incident, was (1) a student enrolled at the University or (2) an employee of the University.

Mandatory Reporters must file a report regardless of how the information comes to their attention – including but not limited to face-to-face conversations, a written class assignment or paper, class discussion, email, text, or social media post. Although Mandatory Reporters must file a report, in most instances, you will be able to control how the report is handled, including whether or not to pursue a formal investigation. The University's goal is to make sure you are aware of the range of options available to you and to ensure access to the resources you need.

Students wishing to discuss concerns in a confidential setting are encouraged to make an appointment with Counseling and Psychological Services (CAPS).

Students can learn more about filing a report, accessing supportive resources, and navigating the Title IX investigation and resolution process on the University's Title IX webpage.

Statement on Mental Health and Wellness

Texas A&M University recognizes that mental health and wellness are critical factors that influence a student's academic success and overall wellbeing. Students are encouraged to engage in proper self-care by utilizing the resources and services available from Counseling & Psychological Services (CAPS). Students who need someone to talk to can call the TAMU Helpline (979-845-2700) from 4:00 p.m. to 8:00 a.m. weekdays and 24 hours on weekends. 24-hour emergency help is also available through the National Suicide Prevention Hotline (800-273-8255) or at suicidepreventionlifeline.org.